



National Institute of Technology Calicut
Department of Computer Science and Engineering
MID TEST-1

Subject Code & Title : CS4037D Cloud Computing
Semester : VII
Max. Marks : 20

Degree : B.Tech.
Date : 06.10.2023
Duration : 2 Hours

Instructions:

1. Ensure your answers are precise and relevant. Irrelevant responses may result in negative marks.
2. There will be no extra sheets provided; manage your space effectively and maintain clear handwriting to accommodate all your answers within the given answer booklet.

Answer all the Questions

1. Consider a large enterprise with hundreds of physical servers deployed in a data center. These servers host critical services used by employees across various organizations. The enterprise is looking to optimize its server infrastructure, enhance resource utilization, improve scalability and flexibility, and ensure robust security measures for protecting sensitive data. In the given scenario, which type of hypervisor would be more suitable for the enterprise's server virtualization needs? Provide your choice and justify your answer based on the specific requirements of the enterprise. **[2 Marks]**
2. Explain the journey of a network packet from the Network Interface Card (NIC) to the application running on a Virtual Machine (VM) in a server. Discuss the various stages it traverses, highlighting the additional overheads it faces compared to running the same application on a bare-metal machine. Additionally, explain how these overheads impact the overall performance of the running application. **[2 Marks]**
3. What is the primary characteristic that distinguishes a public cloud from a private cloud? **[1 Mark]**
4. Fill in the blanks with the appropriate cloud service models (PaaS, SaaS, IaaS) for the given descriptions:
[1.5 Marks]
 - (a) Virtualized hardware resources for users to run their own virtual machines: - - - - -
 - (b) Enables developers to build, deploy, and manage applications without worrying about the underlying infrastructure: - - - - -
 - (c) Delivers software applications over the internet, typically on a subscription basis: - - - - -
5. Why is it essential to carefully select a server when deploying VMs in a data center with hundreds of servers? Additionally, discuss the trade-off between VM performance and energy consumption, explaining the factors that influence this balance. **[1 Mark]**
6. What is the difference between horizontal and vertical VM scaling, and what are the trade-offs associated with each approach? **[1 Mark]**
7. Describe your understanding of different network modes, such as NAT, Bridged, and Host-Only, in Oracle VirtualBox, as part of your hands-on assignment experience. Explain how these modes function in virtualized environments, highlighting their key features and use cases. **[1.5 Marks]**
8. Outline the advantages and limitations of high-performance packet processing techniques such as DPDK, eBPF, XDP, and SmartNICs. **[1 Mark]**

9. The VMware ESX memory management paper was written for shadow page tables, and describe the utility of the following main ideas: **[3 Marks]**

- (a) Idle memory tax
- (b) Ballooning
- (c) Content based page sharing

Which of these ideas are still applicable when using nested page tables? Justify your answer.

10. What are the important parameters that affect VM migration performance? Explain in detail about under which workload scenario the specific migration technique works better. **[3 Marks]**

11. OS use privileged instructions to manage hardware resources like page tables and I/O devices. When executed with a VM on a modern hypervisor providing full virtualization, what happens when a guest executes privileged instructions of this sort. Explain your answer. **[3 Marks]**

*****ALL THE BEST*****